

- ✓ Build a Docker image
- ✓ Push image to GHCR
- ✓ Pull image from GHCR
- ✓ Understand authentication

Create GitHub Personal Access Token (PAT)

Step 1 → Open

Go to:

GitHub → Settings → Developer Settings → Personal Access Tokens

Direct path: <https://github.com/settings/tokens>

Step 2 → Generate Token

Choose:

Classic Token

Enable permissions:

- ✓ write:packages
- ✓ read:packages
- ✓ delete:packages (*optional*)

Step 3 → Copy Token

Copy immediately.

You will NOT see it again.

Tip:

PAT = Password for Docker, not GitHub password

2 Create Simple Demo Project

Create Folder

```
mkdir ghcr-demo
```

```
cd ghcr-demo
```

Create Dockerfile

Create file:

Dockerfile

Paste This (Safest Beginner Image)

FROM nginx:alpine

COPY index.html /usr/share/nginx/html/index.html

Create index.html

Create file:

index.html

Paste:

<h1>GHCR Demo Success </h1>

3 Build Docker Image

Now build locally.

Command

Replace yourusername

docker build -t ghcr.io/yourusername/ghcr-demo .

Example:

docker build -t ghcr.io/harpreetkaur/ghcr-demo .

4 Login to GHCR

Critical authentication step.

Command

docker login ghcr.io

Enter:

✓ Username → GitHub username

✓ Password → **PAT token**

NOT GitHub password

5 Push Image to GHCR

Command

```
docker push ghcr.io/yourusername/ghcr-demo
```

What Happens Now?

Local image → Uploaded → GitHub registry

6 Verify on GitHub

Go to:

GitHub → Profile → Packages

You will see:

- ✓ Image name
- ✓ Visibility
- ✓ Version

7 Pull Image Back (Proof of Success)

Delete local image

Remove Image

```
docker rmi ghcr.io/yourusername/ghcr-demo
```

Pull Again

```
docker pull ghcr.io/yourusername/ghcr-demo
```

8 Run Container (Final Visual Proof)

Command

```
docker run -p 8080:80 ghcr.io/yourusername/ghcr-demo
```

Open browser:

<http://localhost:8080>